

Coding & Solution

Date	
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	5 marks

Solution Summary

Repository Link / URL

- Paste your GitHub repository link after uploading your project.

Programming Language(s)

- Python
- HTML
- CSS
- JavaScript

Framework(s) Used

- Flask
- Scikit-learn
- Pandas
- NumPy

Key Features Implemented

- Crop recommendation using machine learning
- Soil analysis
- Weather-based crop prediction
- Agricultural production optimization
- User-friendly web interface

Pending / Incomplete Features

- Real-time weather API integration
- Mobile application
- Multi-language support

Setup / Run Instructions

1. Clone the GitHub repository.
2. Install the required Python libraries.
3. Run app.py.
4. Open the application in a web browser.
5. Enter soil and weather data to receive crop recommendations.

Code Quality Checklist

S.No	Criteria	Status
1.	Code is modular and organized into functions/classes	Yes
2.	Meaningful variable and function names are used	Yes
3.	Code includes comments/documentation where necessary	Yes
4.	Error handling is implemented for critical operations	Yes
5.	The application runs without critical errors	Yes
6.	Code is committed to a version control repository	Yes

Additional Notes / Comments: The project follows clean coding practices with a modular structure, readable code, and reusable components